Experiment No. 3: Bottle Filling Plant Using Relay System

1. Aim

To interface suitable sensors and actuators with DAQ and demonstrate a bottle filling plant using a relay system.

2. Objectives

1. To study the design of a bottle filling plant.
2. To verify the operation of the plant.
3. To study the construction and working of a relay.

3. Apparatus

- Bottle filling plant trainer model
- Proximity sensor
- DC motor
- HMI
- Relay switches
- Pump and limit switch

4. Introduction

A bottle filling plant is used to automatically fill liquids like water and beverages into bottles. It uses a conveyor, sensors, motor, pump and relay system to perform the filling operation.

5. Process

1. Place the bottle on the conveyor.
2. Start the conveyor motor.
3. The proximity sensor detects the bottle and stops the motor.
4. Start the pump and fill the bottle.
5. The limit switch detects the required liquid level.
6. Stop the pump and restart the conveyor motor.
7. Collect the filled bottle at the end.

6. Advantages

1. Saves time and labour.
2. Reduces liquid wastage.
3. Provides accurate filling.
4. Increases production.
5. Reduces human errors.

7. Disadvantages

1. Initial cost is high.
2. Requires regular maintenance.
3. Depends on electrical power.
4. Sensor or relay failure can stop the system.

8. Applications

1. Mineral water industries.
2. Beverage industries.
3. Pharmaceutical industries.
4. Food processing industries.
5. Chemical industries.

9. Conclusion

The bottle filling plant using a relay system was successfully demonstrated. We studied the working of sensors, actuators, relays and the automatic bottle filling process.

If you want, I can:

- Describe how the relay system controls the bottle filling process step-by-step
- List safety precautions and troubleshooting tips for the bottle filling plant experiment
- Explain how the proximity sensor and limit switch function in the relay-controlled filling plant

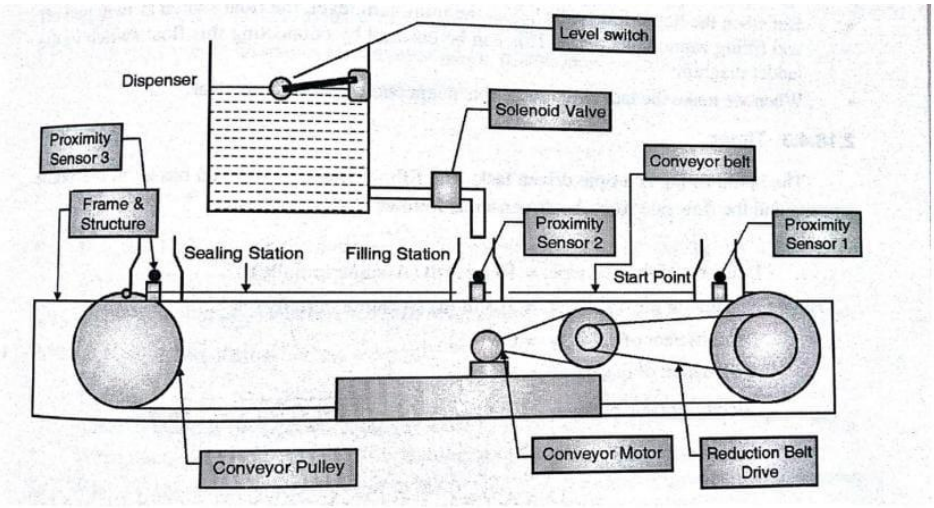


Fig. : Bottle filling station.

Experiment: Interfacing PLC with HMI

1. Aim

To interface a PLC with HMI and monitor and control process variables in real time.

2. Objectives

1. To study the working of PLC and HMI.
2. To establish communication between PLC and HMI.
3. To monitor real-time data.
4. To control outputs using HMI.

3. Apparatus

- PLC
- HMI panel
- Communication cable
- PLC programming software
- HMI configuration software
- Power supply
- Indicator lamp

4. Introduction

HMI (Human Machine Interface) is used to communicate between humans and machines. It displays process information and allows operators to control machines through a screen.

5. Process

1. Connect the PLC to the power supply.
2. Connect PLC and HMI using a communication cable.
3. Create and download the PLC program.
4. Configure the HMI screen and tags.
5. Start the system using the HMI start button.
6. Observe motor and pump status.
7. Press the stop button to stop the output.
8. Verify the system operation.

6. Advantages

1. Easy machine control.
2. Real-time monitoring.

3. Saves time and labour.
4. Reduces human errors.
5. Improves productivity.

7. Disadvantages

1. High initial cost.
2. Requires skilled operators.
3. Depends on electrical power.
4. Communication failure may affect operation.

8. Applications

1. Manufacturing industries.
2. Process industries.
3. Food and beverage industries.
4. Water treatment plants.
5. Automotive industries.

9. Conclusion

The PLC and HMI interfacing experiment was successfully performed. We studied real-time monitoring, communication and machine control using HMI.

If you want, I can:

- Explain how the relay system controls the bottle filling process step-by-step
- Describe the communication protocols used between PLC and HMI in the interfacing experiment
- Suggest improvements or automation upgrades to the bottle filling plant using modern sensors or IoT

5. Block Diagram

